MODULE 04

Machine Learning

A concise 10-page starter workbook for Data Science students

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Learning goal

This workbook introduces the core ideas in machine learning, with a topic-by-topic study path, practical checkpoints, and reflection prompts. Use it as a starting point and supplement it with your class notes and recommended references.

Note: These are sample learning materials created for this demo portal, not official university course notes.

Supervised vs Unsupervised Learning

Concept overview

In this section, focus on the key principles behind **supervised vs unsupervised learning**. Start by defining the problem, identifying the data or assumptions involved, and explaining why the concept matters in a real data science workflow. Keep a short glossary of unfamiliar terms as you study.

A practical way to learn

1. Write the idea in your own words.
2. Work through one small example by hand.
3. Try the same example with a real or synthetic dataset.
4. Record what changed when you adjusted an assumption or parameter.

Checkpoint

Explain supervised vs unsupervised learning to a classmate in three sentences. Then list one situation where it would be useful, one limitation to remember, and one question you still have.

Study notes	
Key term	My explanation
Example	What did I observe?
Open question	What should I investigate next?

Data Preparation and Feature Engineering

Concept overview

In this section, focus on the key principles behind **data preparation and feature engineering**. Start by defining the problem, identifying the data or assumptions involved, and explaining why the concept matters in a real data science workflow. Keep a short glossary of unfamiliar terms as you study.

A practical way to learn

1. Write the idea in your own words.
2. Work through one small example by hand.
3. Try the same example with a real or synthetic dataset.
4. Record what changed when you adjusted an assumption or parameter.

Checkpoint

Explain data preparation and feature engineering to a classmate in three sentences. Then list one situation where it would be useful, one limitation to remember, and one question you still have.

Study notes	
Key term	My explanation
Example	What did I observe?
Open question	What should I investigate next?

Train, Validation, and Test Sets

Concept overview

In this section, focus on the key principles behind **train, validation, and test sets**. Start by defining the problem, identifying the data or assumptions involved, and explaining why the concept matters in a real data science workflow. Keep a short glossary of unfamiliar terms as you study.

A practical way to learn

1. Write the idea in your own words.
2. Work through one small example by hand.
3. Try the same example with a real or synthetic dataset.
4. Record what changed when you adjusted an assumption or parameter.

Checkpoint

Explain train, validation, and test sets to a classmate in three sentences. Then list one situation where it would be useful, one limitation to remember, and one question you still have.

Study notes	
Key term	My explanation
Example	What did I observe?
Open question	What should I investigate next?

Linear Regression

Concept overview

In this section, focus on the key principles behind **linear regression**. Start by defining the problem, identifying the data or assumptions involved, and explaining why the concept matters in a real data science workflow. Keep a short glossary of unfamiliar terms as you study.

A practical way to learn

1. Write the idea in your own words.
2. Work through one small example by hand.
3. Try the same example with a real or synthetic dataset.
4. Record what changed when you adjusted an assumption or parameter.

Checkpoint

Explain linear regression to a classmate in three sentences. Then list one situation where it would be useful, one limitation to remember, and one question you still have.

Study notes	
Key term	My explanation
Example	What did I observe?
Open question	What should I investigate next?

Classification with Logistic Regression

Concept overview

In this section, focus on the key principles behind **classification with logistic regression**. Start by defining the problem, identifying the data or assumptions involved, and explaining why the concept matters in a real data science workflow. Keep a short glossary of unfamiliar terms as you study.

A practical way to learn

1. Write the idea in your own words.
2. Work through one small example by hand.
3. Try the same example with a real or synthetic dataset.
4. Record what changed when you adjusted an assumption or parameter.

Checkpoint

Explain classification with logistic regression to a classmate in three sentences. Then list one situation where it would be useful, one limitation to remember, and one question you still have.

Study notes	
Key term	My explanation
Example	What did I observe?
Open question	What should I investigate next?

Decision Trees and Random Forests

Concept overview

In this section, focus on the key principles behind **decision trees and random forests**. Start by defining the problem, identifying the data or assumptions involved, and explaining why the concept matters in a real data science workflow. Keep a short glossary of unfamiliar terms as you study.

A practical way to learn

1. Write the idea in your own words.
2. Work through one small example by hand.
3. Try the same example with a real or synthetic dataset.
4. Record what changed when you adjusted an assumption or parameter.

Checkpoint

Explain decision trees and random forests to a classmate in three sentences. Then list one situation where it would be useful, one limitation to remember, and one question you still have.

Study notes	
Key term	My explanation
Example	What did I observe?
Open question	What should I investigate next?

Clustering with K-Means

Concept overview

In this section, focus on the key principles behind **clustering with k-means**. Start by defining the problem, identifying the data or assumptions involved, and explaining why the concept matters in a real data science workflow. Keep a short glossary of unfamiliar terms as you study.

A practical way to learn

1. Write the idea in your own words.
2. Work through one small example by hand.
3. Try the same example with a real or synthetic dataset.
4. Record what changed when you adjusted an assumption or parameter.

Checkpoint

Explain clustering with k-means to a classmate in three sentences. Then list one situation where it would be useful, one limitation to remember, and one question you still have.

Study notes	
Key term	My explanation
Example	What did I observe?
Open question	What should I investigate next?

Model Evaluation Metrics

Concept overview

In this section, focus on the key principles behind **model evaluation metrics**. Start by defining the problem, identifying the data or assumptions involved, and explaining why the concept matters in a real data science workflow. Keep a short glossary of unfamiliar terms as you study.

A practical way to learn

1. Write the idea in your own words.
2. Work through one small example by hand.
3. Try the same example with a real or synthetic dataset.
4. Record what changed when you adjusted an assumption or parameter.

Checkpoint

Explain model evaluation metrics to a classmate in three sentences. Then list one situation where it would be useful, one limitation to remember, and one question you still have.

Study notes	
Key term	My explanation
Example	What did I observe?
Open question	What should I investigate next?

Overfitting, Regularization, and Model Selection

Concept overview

In this section, focus on the key principles behind **overfitting, regularization, and model selection**. Start by defining the problem, identifying the data or assumptions involved, and explaining why the concept matters in a real data science workflow. Keep a short glossary of unfamiliar terms as you study.

A practical way to learn

1. Write the idea in your own words.
2. Work through one small example by hand.
3. Try the same example with a real or synthetic dataset.
4. Record what changed when you adjusted an assumption or parameter.

Checkpoint

Explain overfitting, regularization, and model selection to a classmate in three sentences. Then list one situation where it would be useful, one limitation to remember, and one question you still have.

Study notes	
Key term	My explanation
Example	What did I observe?
Open question	What should I investigate next?